

Describe the relationship between perfect square numbers and square roots, and use squares of numbers and square roots of perfect square numbers to solve problems

Learning intention

- describe the relationship between perfect square numbers and square roots, and use squares of numbers and square roots of perfect square numbers to solve problems.

Teach and model

- generating a list of perfect square numbers and describing any emerging patterns.
- determining between which 2 consecutive natural numbers the square root of a given number lies.
- using the square and square root notation, and the distributive property and area diagrams to calculate the squares of two-digit numbers.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.